ST 3189

Machine Learning Coursework

Margaret Janelle Kamto
220657356

Table of Contents

1. Introduction **2**
2. Dataset and Exploratory Data Analysis (EDA) **2**
2.1. Unsupervised Learning (Breast Cancer) 2
2.2. Regression Dataset (Bank Marketing) 3
2.3. Classification Dataset (Bank Marketing) 4
3. Methodology **4**
3.1. Unsupervised Learning (Breast Cancer) 4
3.1.1. Principal Component Analysis (PCA) 4
3.1.2. Clustering Methods 5
3.2. Regression (Bank Marketing) 5
3.3. Classification (Bank Marketing) 6
4. Result and Discussion **6**
4.1. Unsupervised Learning (Breast Cancer) 6
4.1.1. Principal Component Analysis (PCA) 6
4.1.2. K-Means Clustering 7
4.1.3. Hierarchical Clustering 8
4.2. Regression (Bank Marketing) 8
4.3. Classification (Bank Marketing) 10
5. Conclusion **11**
6. References **12**

1. Introduction

Machine learning is a branch of artificial intelligence, which makes predictions by using algorithms that learn from data. This project explores both unsupervised and supervised machine learning techniques on real world datasets to discover patterns and make predictions. This study begins with unsupervised learning applying clustering techniques on breast cancer dataset to identify hidden patterns in patient records. The second and third sections are application of supervised learning on bank marketing dataset, using regression models to predict customer call durations and classification models to forecast customer behaviour on subscription to a term deposit. This project applies suitable machine learning models for each task, evaluates performance, and interprets results in order to gain meaningful insights.

2. Dataset and Exploratory Data Analysis (EDA)

2.1. Unsupervised Learning (Breast Cancer)

The unsupervised learning part uses the Breast Cancer Gene Expression Profiles (METABRIC) dataset from Kaggle. The original dataset contains 1,980 patient records, while this analysis uses a version with 1,904 entries with both clinical and mutation data available. There are 5 selected features in this task, are tumor_size, age_at_diagnosis, mutation_count, lymph_nodes_examined_positive, and nottingham_pragnostic_index, that are chosen due to clinical importance and continuous data type, making them ideal candidates suitable for clustering algorithm like K-Means. The missing values check showed only mutation_count and tumor_size has 65 missing values in total out of 1,904 rows, representing about 3.4% which is small proportion of the dataset containing missing values, thus dropna() is used to remove them, maintaining data quality with minimal impact on sample size.

Selected features histograms show age_at_diagnosis is normally distributed, mutation_count, tumor_size, and lymph_nodes_examined_positive are right-skewed, while nottingham_pragnostic_index shows grouped values due to clinical scoring. These distribution patterns help the clustering process. Correlation heatmap of selected features shows the strongest correlation (0.56) between lymph nodes and prognostic index, indicating that more lymph node spread is linked to poorer prognosis. Tumor size and prognostic index have weak correlation (0.28), meaning bigger tumors slightly increase prognosis index. Other features have no extreme correlations, suggesting they are independent and have no redundant features which is ideal for unsupervised learning.

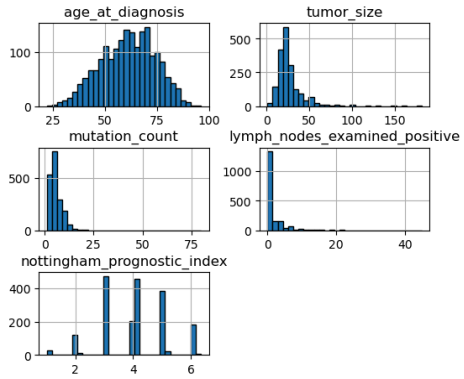

Figure 1: Histogram Distributions of Unsupervised Selected Features

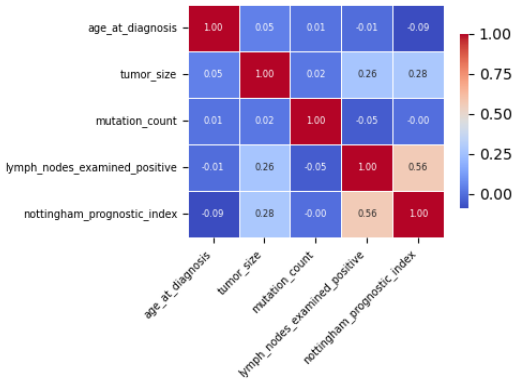

Figure 2: Correlation Heatmap of Unsupervised Selected Features

2.2. Regression Dataset (Bank Marketing)

Regression and classification tasks use the same dataset which is the Bank Marketing dataset from UCI Machine Learning Repository. It contains 41,188 entries and 21 features related to bank customers and marketing campaign information, consisting of both customer demographics and economic indicators.

The target variable (y) is duration, representing call length in second for the last call with a client. It is continuous and offers valuable insights about customer engagement, making it an ideal target for regression analysis. Nine numerical features which potentially impact call length are selected, such as campaign, age, duration, previous, pdays, emp.var.rate, cons.conf.idx, cons.price.idx, nr.employed, euribor3m. The missing values check shows no missing values, hence no data imputation or removal are needed. Certain features show strong right skewness, leading to log and square root transformation. The final preprocessing transformations included normalizing campaign using Min-Max scaling, squaring duration_squared, and applying binary encoding previous and pdays for better feature distribution and model performance.

Correlation heatmap shows weak linear relationships between duration and other features, with the strongest correlation at |-0.15| with a campaign. High multicollinearity is also observed among the economic indicators and outcome variables. To address weak correlations between duration and selected features, improve data representation, and model's ability to identify patterns, feature engineering is applied to create new features capturing non-linear effects.

Table 1: Bank Marketing Dataset Feature Engineering Table

Feature	Formula	Interpretation
employment_contact	nr.employed * contact	Check the impact of employment rate on contact type (telephone vs. cellular), enabling modeling joint influence
duration_squred	$duration^2$	Reduce skewness in the target variable, helping the models capture non-linear relationships
pdays_squared	$pdays^2$	Capture the non-linear effect between the recency of last contact and call length
economic_cluster	emp.var.rate + cons.price.idx + euribor3m	Summarize economic indicator, grouping customers based on economic conditions

Final histograms after transformation and feature engineering indicate that duration_squared becomes more normally distributed, binary-encoded features which are pdays and pdays_previous present clear 0-1 separation. Campaign and employment_contact still imbalance yet contains meaningful structure for modeling. Campaign skewed distribution reflects real marketing behaviour. The sharp concentration and no uniformly distributed employment_contact reflects how employment levels correlates with communication channels. Although these two features are not evenly spread, they still provide meaningful insights for the regression model.

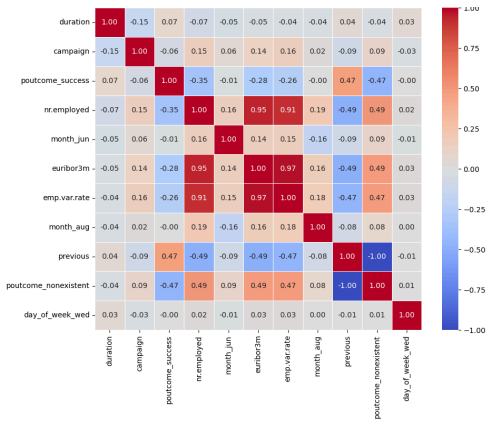

Figure 3: Histogram Distributions of Regression & CLassification Selected Features

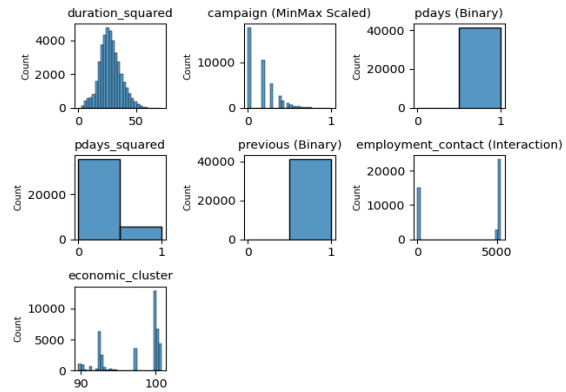

Figure 4: Correlation Heatmap of Regression & Classification Selected Features

2.3. Classification Dataset (Bank Marketing)

Dataset used in classification is the same as in Section 2.2, which is the Bank Marketing dataset. The objective is to predict customers' subscription to a term deposit. The target variable (y) in this task is binarised, where “yes” is encoded as 1 and “no” as 0. It is selected as it reflects an important business outcome measuring campaign success. Hereunder figure 5 shows the imbalanced dataset that most customers are not subscribing to a term deposit.

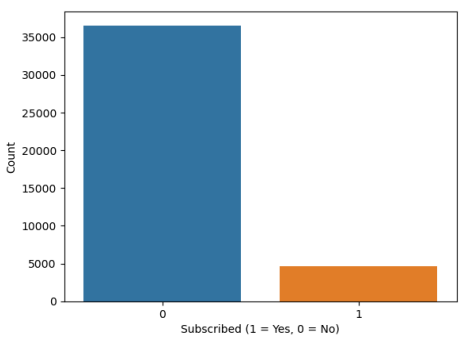

Figure 5: Class Distributions of Term Deposit Subscription

The selected features used for classification are similar with regression tasks, which are pdays_squared, economic_cluster, previous, duration_squared, employment_contact, nr.employed, and cons.price.idx. These variables reflect a combination of behaviour, economic indicator, and campaign related factors predicting subscriptions.

3. Methodology

3.1. Unsupervised Learning (Breast Cancer)

3.1.1. Principal Component Analysis (PCA)

Principal Component Analysis (PCA) is applied before clustering to reduce dimensionality. Selected numerical features are standardized using StandardScaler, ensuring equal contribution. The explained variance ratio and scree plot visualization are used to determine the number of components to retain. Based on the analysis, the first four principal components (PCs) are

selected to capture the majority of the data variance while reducing dimensionality for clustering.

3.1.2. Clustering Methods

Two clustering methods applied in this project are K-Means and Hierarchical Clustering. K-Means clustering is performed on the PCA-reduced dataset to discover potential groupings by using two evaluation methods such as Elbow Method and Silhouette Score, which are implemented to determine the optimal number of clusters (K). The Elbow Method is applied by plotting Within-Cluster Sum of Squares (WCSS) for different K values. From the model's inertia, WCSS measures the total squared distance from each point to its corresponding cluster center. The optimal number of clusters is determined at the "elbow" point where the reduction in WCSS starts to flatten (Kodinariya & Makwana, 2013). The Silhouette Score ranges from -1 to +1, evaluates how well data points fit within their own clusters than that of in other clusters. The higher the values, the better the clustering is (Ankita, 2025). K values ranging from 2 to 10 are tested, with K=2 chosen based on the sharp elbow point and highest silhouette score. Clustering results are later visualized using the first two PCs. Agglomerative hierarchical clustering with Ward's linkage, which minimizes within-cluster variance, is applied to analyze hierarchical grouping structure which is visualized using dendrogram. A significant jump in linkages indicates a natural split in data, where clusters become distinct and well-separated (C, 2023).

3.2. Regression (Bank Marketing)

The objective of the regression task is to predict duration (y) in second for the last call with a client using selected numerical features. Three regression models selected for comparison. Linear Regression as a baseline to evaluate whether a basic model could capture patterns. Random Forest Regression for its capability to capture non-linear patterns and reduce overfitting. XGBoost Regression as a boosting model which performs best for weak correlations between target and selected numerical features. Lasso Regression is not applied as duration has weak correlations so not many redundant features. Random Forest Regression is a better version of Decision Tree Regression, hence no need to do both. Selected features are standardized using StandardScaler. The dataset is split into 80% training and 20% testing to evaluate performance and avoid overfitting. Model performance is assessed using regression evaluation metrics such Mean Absolute Error (MAE), Mean Squared Error (MSE), and the R^2 score based on residual. Residual is the differences or errors between actual and predicted values. MAE measures the average of these residuals. MSE highlights larger errors by squaring residuals, making it more sensitive to prediction inaccuracies. R^2 score reflects how much variances in the target variable explained by the model. The higher the R score, the better the overall fit to the data. Other metrics like Root Mean Squared Error (RMSE) or Mean Absolute Percentage Error (MAPE) are not considered in this project because MAE and MSE already capture errors and sensitivity, whilst R^2 score represents the model's overall ability in capturing trends. These three metrics already provide a comprehensive and interpretable evaluation.

3.3. Classification (Bank Marketing)

The classification focuses on predicting customer subscription to a term deposit (y). Since the target variable is binary outcome (yes/no), it is suitable for classification models. For model training, the same training and testing sets in regression are also applied in classification using 80:20 stratified split to maintain class distribution. Three classification models are chosen for comparison. Logistic Regression as a baseline to provide a benchmark for model performance comparison. Decision Tree Classifier captures non-linear relationships and is interpretable through its tree structure. XGBoost Classifier is a powerful ensemble model which is good in handling complex relationships between features and imbalanced data. These models are selected to represent different complexities while avoiding redundancy. Each offers distinct strength and insightful comparison for customer subscription prediction. To assess classification performance, models are evaluated using accuracy, precision, recall, F1-score, and ROC-AUC. Accuracy gives the overall rate of the correct predictions yet when classes are imbalanced, it can be misleading. Precision reflects how many relevant positive predictions. Higher precision values indicate fewer false positives. Recall represents the model's ability to detect true positives correctly that higher recall scores means fewer false negatives. F1-score is the combination of precision and recall which is suitable when both error types occur. ROC-AUC explains how well the model separates the classes across the threshold that a higher score means better separation. These metrics are selected due to their ability to provide balanced and reliable evaluation particularly in cases where data is imbalanced.

4. Result and Discussion

4.1. Unsupervised Learning (Breast Cancer)

The research question for unsupervised learning task is: Can dimensionality reduction and clustering methods identify distinct patient groups based on genetic and clinical characteristics in the breast cancer dataset?

4.1.1. Principal Component Analysis (PCA)

Based on the scree plot and explained variance analysis, the first four principal components (PCs) captured 91.5% of the dataset's variance, with individual explained variance such as PC1=35.2%, PC2=20.6%, PC3=20.1%, PC4=15.6%. PC5 contributed only 8.5% of the total variance, contributing relatively low information beyond the first four. As illustrated in the scree plot, there is a steep rise in cumulative variance across the first four PCs and starts to flatten after the fourth component. The "elbow" at PC4 suggests to retain only the top four as an optimal balance between dimensionality reduction and variance retention, which simplify the dataset while maintaining key variance for effective clustering analysis.

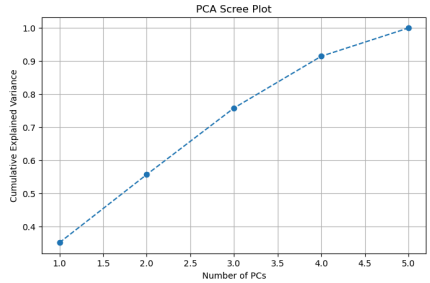

Figure 6: PCA Scree Plot Showing Cumulative Explained Variance

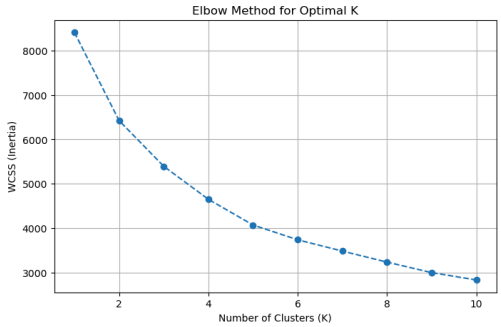

Figure 7: Elbow Method Showing Optimal K

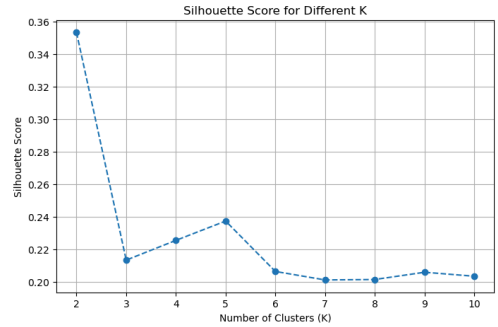

Figure 8: Silhouette Score for Different K

4.1.2. K-Means Clustering

The Elbow Method shows a distinct bend at K=2, indicating it as the optimal number of clusters. It aligns with Kodinariya and Makwana (2013), explaining the elbow as the optimal number of clusters determined at the "elbow" point where the reduction rate begins to flatten, implying a balance between model complexity and cohesion. The Silhouette Score reaches a maximum value of 0.354 at K=2 and decreases thereafter, implying that the two-cluster is the best defined grouping. According to Ankita (2025), a higher silhouette score close to +1 is a good clustering that data points in the same cluster should be grouped together and distinctly separated from other clusters. Although the value near 0 suggests moderate clustering, this is still the highest score across all tested K values. This is supported by the Calinski-Harabasz score at K=2 which is 568.18, validating the compact and well separated clusters. Cluster 0 (n=374) represents patients with more severe clinical features such as larger tumor sizes, higher Nottingham Prognostic Index (NPI), and higher lymph node involvement. Cluster 1 (n=1465) consists of patients with better clinical profiles, such as smaller tumor sizes and lower prognostic indicators. These differences are shown in hereunder figures. The two clusters are also clearly separated as shown in the PCA scatter plot.

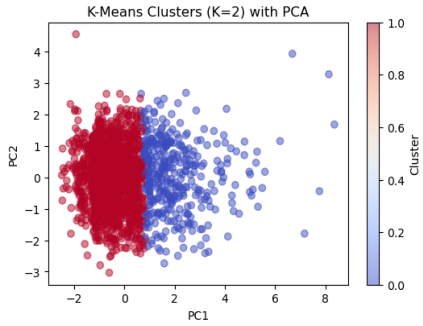

Figure 9: Scatter Plot with K=2

Figure 11: Cluster Size Distribution (K=2)

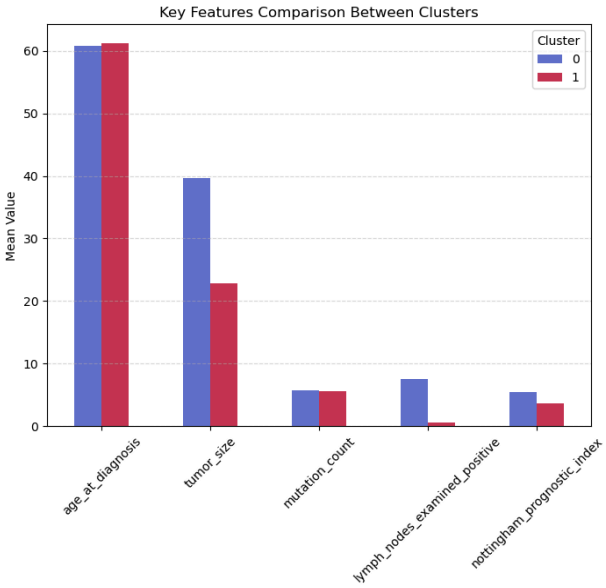

Figure 10: Mean Features Comparison Between Clusters (K=2)

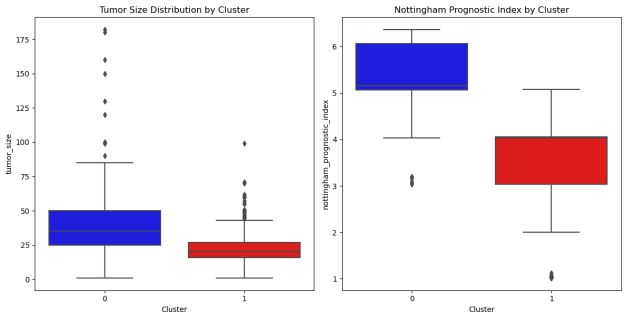

Figure 12: Boxplots of Tumor Size and Prognostic Index (K=2)

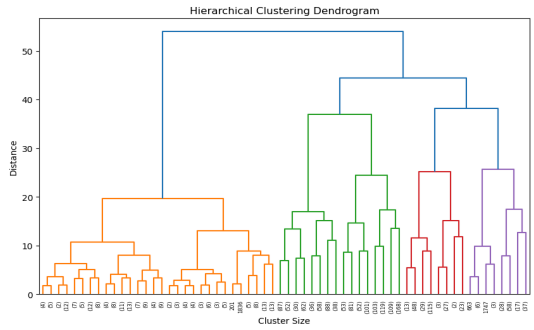

Figure 13: Hierarchical Clustering Dendrogram (Linkage Method)

4.1.3. Hierarchical Clustering

Hierarchical clustering with Ward linkage is performed to the same PCA-reduced data. The dendrogram shows a distinct two main branches, implying K=2 is an appropriate number of clusters. While the clusters are not visually well separated compared to K-Means, both methods identify two meaningful groupings, validating the overall clustering results. This result is supported by Priya C (2023), suggesting the largest vertical linkages is the optimal point to split the data into distinct clusters. Despite moderate silhouette score below 0.5, hierarchical clustering further validates K-Means result at K=2, indicating a consistent underlying pattern in the data from both clustering methods.

4.2. Regression (Bank Marketing)

The research question for the regression task is: How accurately the duration of the most recent client predicted based on campaign-related features and economic indicators in the bank marketing dataset? An 80:20 train-test split applied to evaluate model performance, resulting in 32,940 training samples and 8,236 test samples. Among the three tested models, Random Forest Regression outperforms other models with the highest R^2 score (0.999995), MAE (0.000056), and MSE (0.000004). This model has the highest R^2 score, implying the model explains the most variance in the target variable; the lowest MAE, meaning the most accurate predictions; and the lowest MSE, indicating the least variance in errors prediction. Followed by XGBoost Regression with comparable R^2 score (0.998782), while Linear Regression underperforms due to its poorer fit and inability to capture non-linear patterns.

Table 2: Regression Models Performance Metrics

	Model	MAE	MSE	R^2 Score
0	Linear Regression	0.086427	0.018607	0.977778
1	Random Forest	0.000056	0.000004	0.999995
2	XGBoost	0.006432	0.001020	0.998782

Residual analysis supported these metric results. Based on the residual histogram, XGBoost and Random Forest are tightly clustered around zero, suggesting minimal prediction error and better fit. The visualizations of Random Forest bar is almost invisible as it is so narrow and close to zero. In contrast, Linear Regression is left-skewed and shows greater residual spread, indicating a non-normality and less precise predictions (OriginLab, 2018). In residuals vs. predicted values plot, Linear Regression shows curvilinear pattern which indicates there is a missing higher-order term. The model fails to capture non-linear patterns, suggesting poor fit.

However, both Random Forest and XGBoost residuals are generally centred around zero without any uneven residuals spread or curvature, suggesting these models satisfy the assumptions of constant variance and linearity. Random Forest has more concentrated and tighter residuals than that of XGBoost which produces more spread residuals yet still symmetrical. These interpretations align with Minitab's explanation, stating that the absence of uneven distribution of residuals along the predicted values and curvilinear pattern suggest a good-fitting model (Minitab, 2024). The Q-Q plots also validate that residuals from Random Forest and XGBoost closely approximated a normal distribution with Random Forest more closely following the normal line, while XGBoost displays light tails yet remained centered and symmetrical. In contrast, Linear Regression shows an S-shaped curve, implying there is skewness or non-normal distributions (Waples, 2024).

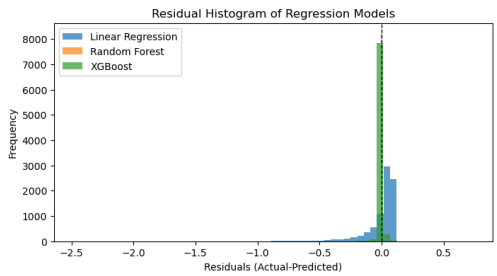

Figure 14: Residual Histogram of Regression Models

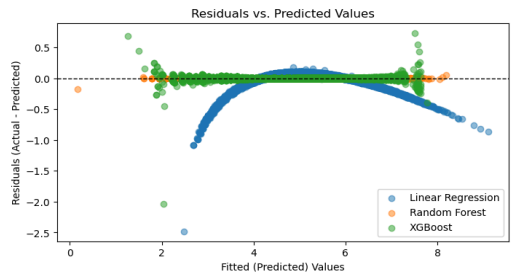

Figure 15: Residual vs. Fitted Values Plot of Regression Models

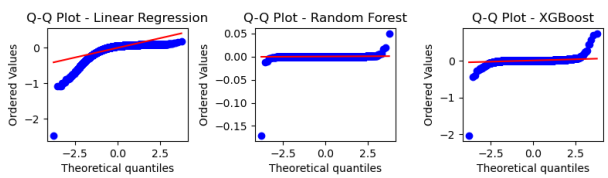

Figure 16: Q-Q Plot of Regression Models

The feature importance visualization shows that duration_squared is the top predictor in both Random Forest and XGBoost. Other key features like cons.price.idx, economic_cluster, and nr.employed consistently show high importance in both models. Whereas employment_contact, previous, and pdays_squared are relatively less important. Variable pdays_squared contributes more in XGBoost compared to Random Forest, implying XGBoost might capture more complex interactions related to this variable. In conclusion, XGBoost is the most accurate model in predicting the duration of the most recent client contact.

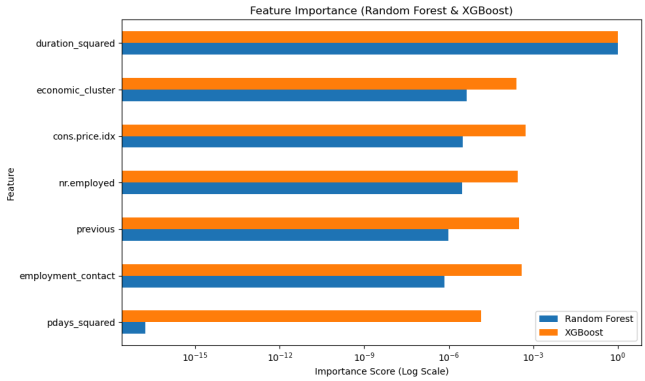

Figure 17: Feature Importance of Random Forest & XGBoost

4.3. Classification (Bank Marketing)

The research question for classification task is: How effectively can demographic and campaign-related features predict customer subscription to a term deposit? Using an 80:20 train-test split, models are trained on 32,904 samples for training and 8,236 samples for testing. These three classification models are evaluated based on a highly imbalanced dataset in which the majority of the customers did not subscribe to a term deposit. XGBoost outperformed the other models on all evaluation metrics Accuracy (91.36%), Precision (63.08%), Recall (56.14%), F1 (59.41%), ROC-AUC (94.30%). Although Logistic Regression has strong precision (90.42%) and ROC-AUC (91.97%), it struggles with low recall (36.85%), meaning that many subscribers are not correctly identified. Decision Tree offers moderate performance yet the lowest ROC-AUC (73.52%), indicating weaker class separation compared to other models.

Table 3: Classification Models Performance Metrics

	Model	Accuracy	Precision	Recall	F1 Score	ROC-AUC
0	Logistic Regression	0.904201	0.627523	0.368534	0.464358	0.919715
1	Decision Tree	0.893638	0.531175	0.477371	0.502838	0.735189
2	XGBoost	0.913550	0.630751	0.561422	0.594071	0.943042

Table 4: Confusion Matrix of Classification Models

	Model	True Negatives (TN)	False Positives (FP)	False Negatives (FN)	True Positives (TP)
0	Logistic Regression	7105	203	586	342
1	Decision Tree	6917	391	485	443
2	XGBoost	7003	305	407	521

These findings are supported by confusion matrices. XGBoost has the highest true positives (521) and the lowest false negatives (407), suggesting it is the most accurate at identifying actual subscribers. Logistic Regression has the lowest false positives, making few incorrect "yes" predictions and highest false negatives (586), implying it had poor call and missed many actual subscribers. According to Vidhi Chugh (2024), a high-performing model on the ROC curve will bend toward the top left corner, where the false positives are minimized and true positives are maximized. The ROC curve shows that XGBoost reached the highest AUC score (0.94) close to 1, which further confirms XGBoost has excellent performance with strong classification ability to separate classes. Its precision-recall curve shows stable and high increase throughout recall values, which validates more its consistency in identifying actual subscribers. Decision Tree shows nr.employed and duration_squared as primary predictors, validating their role in predicting subscriptions. Overall, XGBoost is the most effective model in predicting customer predictions which balances recall and precision, showing its superior overall performance.

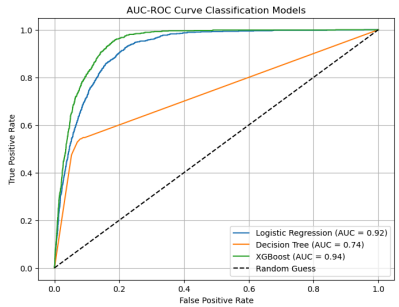

Figure 18: AUC-ROC Curve of Classification Models

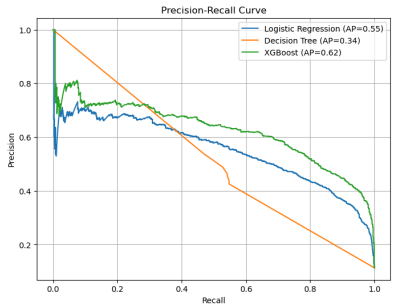

Figure 19: Precision-Recall Curve of Classification Models

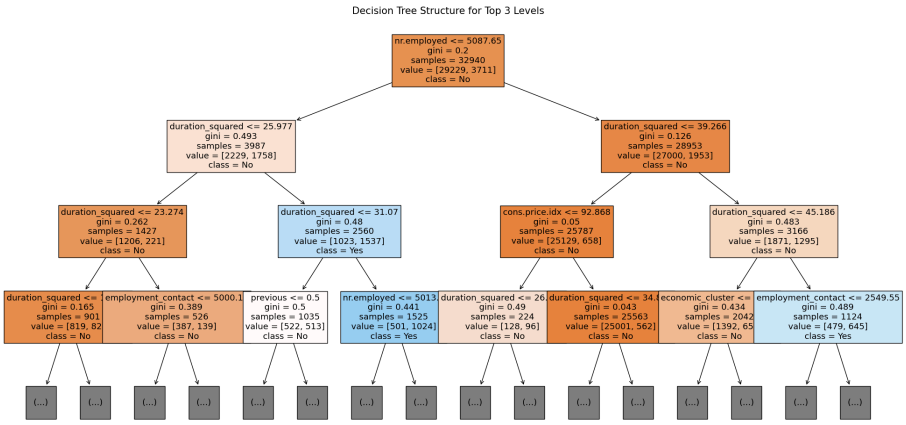

Figure 20: Decision Tree for Top 3 Levels

5. Conclusion

This project implemented three core machine learning approaches which are unsupervised learning, regression, and classification with real world datasets application. Unsupervised learning task successfully identify two different patient groups in the breast cancer (METABRIC) dataset using Principal Component Analysis (PCA) and clustering method. PCA reduced dimensionality while retaining more than 90% of the variance. Both K-means and hierarchical clustering produces consistent and interpretable groupings of two distinct clusters. The research question for the regression task evaluates how accurately the duration of the most recent client predicted based on campaign-related features and economic indicators. Three regression models are compared, with XGBoost Regression outperformed Linear Regression and Random Forest Regression by producing the highest R^2 score and the lowest MAE and MSE, suggesting the most accurate model in predicting strength of boosting techniques in modeling complex data patterns and weak correlation between target variable (y) and selected features. These results are further validated by some regression visualizations. Classification analysis focuses on how effectively demographic and campaign-related features predict customer subscription to a term deposit. XGBoost classifier has the best performance across all metrics, outperforming Decision Tree models and Logistic Regression. The model's ability in handling imbalance data and enhancing prediction accuracy is the key to its effectiveness. Overall, these findings highlight the significance of selecting suitable models, applying proper processing steps, and critically interpreting model performance using evaluation metrics. XGBoost consistently outperformed baseline line models like Linear Regression and Decision Trees in both regression and classification tasks. Unsupervised learning task highlights how dimensionality reduction and clustering give deeper insight into underlying patterns within complex data.

6. References

Kodinariya, T. M., & Makwana, P. R. (2013). Review on determining number of Cluster in K-Means Clustering. *International Journal of Advance Research in Computer Science and Management Studies*, *1*(6), 3. https://doi.org/https://www.researchgate.net/publication/313554124

Ankita, A. (2025, March 10). *K-means: Getting the optimal number of clusters*. Analytics Vidhya. https://www.analyticsvidhya.com/blog/2021/05/k-mean-getting-the-optimal-number-of-clusters/

C, B. P. (2023, October 6). *Unveiling hidden patterns: An introduction to hierarchical clustering*. KDnuggets. https://www.kdnuggets.com/unveiling-hidden-patterns-an-introduction-to-hierarchical-clustering

OriginLab. (2018). *15.4.4 residual plot analysis*. OriginLab Corporation - Data Analysis and Graphing Software - 2D graphs, 3D graphs, Contour Plots, Statistical Charts, Data Exploration, Statistics, Curve Fitting, Signal Processing, and Peak Analysis. https://www.originlab.com/doc/origin-help/residual-plot-analysis

Minitab. (2024). *Residual plots for fit regression model and linear regression*. https://support.minitab.com/en-us/minitab/help-and-how-to/statistical-modeling/regression/how-to/fit-regression-model/interpret-the-results/all-statistics-and-graphs/residual-plots/

Waples, J. (2024, November 17). *The Q-Q plot: What it means and how to interpret it*. DataCamp. https://www.datacamp.com/tutorial/qq-plot

Chugh, V. (2024, September 10). *AUC and the ROC curve in machine learning*. DataCamp. https://www.datacamp.com/tutorial/auc